

Cyrus Guided Notes: MIT 6.245 Multivariable Control Systems

MIT course pack

2026-05-15

Course source

Official source: <https://ocw.mit.edu/courses/6-245-multivariable-control-systems-spring-2004/>

Why this course is in the system. Graduate control course for MIMO systems, robustness, and frequency-domain design.

Learning output. Keep it in the graduate control chapter instead of forcing it into the first beginner round.

Prerequisite checkpoint

- State space
- Frequency response
- Linear systems

Formula checkpoint

Do not memorize these first. Read each formula as a sentence: what changes, what is observed, what is optimized, and what output it produces.

$$\begin{aligned} \mathbf{y}(s) &= G(s)\mathbf{u}(s) \\ \bar{\sigma}(G(j\omega)) \\ \|T_{zw}\|_{\infty} &< \gamma \\ S &= (I + GK)^{-1}, \quad T = GK(I + GK)^{-1} \end{aligned}$$

Visual page

Draw MIMO as many inputs pushing many outputs, then show singular values as worst-direction gain.

GoodNotes pages

- Page MIT-M001: MIMO transfer matrix

- Page MIT-M002: sensitivity and robustness

Web interaction

A gain-direction widget that rotates an input vector and shows output amplification.

Obsidian and Notion

Layer	Output
Obsidian	MIT Control -> 6.245 multivariable control
Notion	Robust-control expansion queue.

First study move

Open the official source, copy the prerequisite checkpoint into GoodNotes, then write one formula in your own words before watching or reading more.